

In our life we've already come across functions that contain multiple variables. For example, the volume of a cylinder of radius r and height h is given by

$$V(r, h) = \pi r^2 h.$$

If we want to formalize this notion a little more, we can say the following:

Definition. If D is a set of n -tuples of real numbers $(x_1, x_2, \dots, x_n)$, then an $\mathbb{R}$ -**valued function** f on D is a rule that assigns a unique real number for every tuple in its domain $D = D_f = \text{Dom}(f)$. The set of possible outputs is f 's **range**. The x_i are the independent variables and the single, real output is the dependent variable.

Example. Discuss the domains and ranges for the functions

$$g_1(x, y) = \frac{x^2 + y^2}{y} \quad \text{and} \quad g_2(x, y) = (x\sqrt{y}, x + y, y\sqrt{x}).$$

In both of these examples, the domains were subsets of the xy -plane (i.e., subsets of $\mathbb{R}^2$). We can represent these visually using traditional graphing techniques. Let's consider the graph of some bivariate function $z = f(x, y)$:

Recall that for single-variable functions, their domains were often combinations of the 1-dimensional intervals (a, b) , $(a, b]$, $[a, b)$, and $[a, b]$. Let's introduce some definitions to help classify 'intervals' in $\mathbb{R}^2$.

- Interior Point (forming the 'interior')
- Boundary Point (forming the 'boundary')
- Open (has no boundary)
- Closed (contains its boundary)

Example. The simplest 'interval' in $\mathbb{R}^2$ is the unit disc. Consider the following variants of it:

$$S_1 = \{(x, y) : x^2 + y^2 < 1\} \quad S_2 = \{(x, y) : x^2 + y^2 = 1\} \quad S_3 = \{(x, y) : x^2 + y^2 \leq 1\}$$

If a region of the plane lies entirely inside a visible range, then that region is **bounded**—the above discs are examples of bounded sets. If this is not possible, then the set is **unbounded**. Example of unbounded sets are planes and lines.

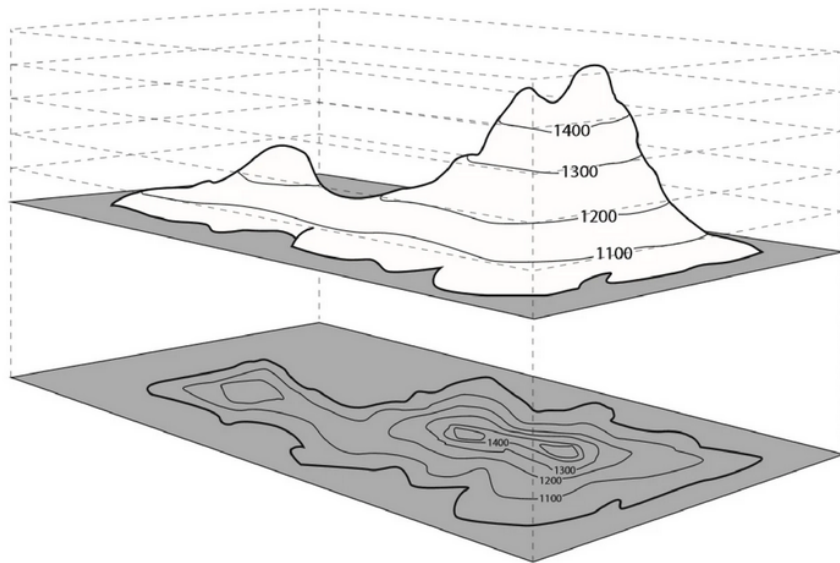
Example. Describe the domain of the function $z = f(x, y) = \sqrt{y - x^2}$.

Now we know how to accurately describe the domain of a multivariate function $z = f(x, y)$. But how do we generate the graph itself? That is, how do we draw its *surface*? We learned in Chapter 11 that oftentimes looking at *slices* is a good idea. We now formalize this concept:

Definition. The set of points in 3-space where a function $z = f(x, y)$ has constant value (i.e., $f(x, y) = c$ for some constant c) is called a **level contour** of f .

Example. Graph $z = f(x, y) = 100 - x^2 - y^2$ and note the level contours at $z = 0$, $z = 51$, and $z = 75$. Then ‘drop’ (or ‘project’) the level curves onto the xy -plane so that they generate regions within the domain of f —these objects are called the **contour levels** (or **level sets**).

Example. The best example of these objects in real-life are maps that contain elevation contours:



Extending into 3-space

Suppose now we have the trivariate function $w = f(x, y, z)$. Graphing these is not possible (we cannot ‘see’ 4-dimensional objects), but much of our machinery still works.

One example is that, in 3-space, the notion of ‘interior’ and ‘boundary’ still makes sense and therefore so does the idea of an ‘open’ and ‘closed’. Consider the following:

(i) $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 < 1\}$

(ii) $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$

(iii) $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1\}$

Given a trivariate function $w = f(x, y, z)$, its domain is now a subset of $\mathbb{R}^3$ and so the level curves are shapes in $\mathbb{R}^3$. Because of this, they are now called **level surfaces**.

Example. Describe the level surfaces of $w = f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$.